

Cancer Statistics, 2004

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ABSTRACT Each year, the American Cancer Society estimates the number of new cancer cases and deaths expected in the United States in the current year and compiles the most recent data on cancer incidence, mortality, and survival rates based on incidence data from the National Cancer Institute and mortality data from the National Center for Health Statistics. Incidence and mortality rates are age standardized to the 2000 US standard million population. A total of 1,368,030 new cancer cases and 563,700 deaths are expected in the United States in 2004. Incidence rates stabilized among men from 1995 through 2000 but continued to increase among females by 0.4% per year from 1987 through 2000. Mortality rates have decreased by 1.5% per year since 1992 among men, but have stabilized from 1998 through 2000 among women. Cancer death rates continued to decrease from the three major cancer sites in men (lung and bronchus, colon and rectum, and prostate) and from female breast and colorectal cancers in women. In analyses by race and ethnicity, African-American men and women have 40% and 20% higher death rates from all cancers combined compared with White men and women, respectively. Cancer incidence and mortality rates are lower in other racial and ethnic groups than in Whites and African Americans for all sites combined and for the four major cancer sites. However, these groups generally have higher rates for stomach, liver, and cervical cancers than do Whites. Furthermore, minority populations are more likely to be diagnosed with advanced stage disease than are Whites. Progress in reducing the burden from cancer can be accelerated by applying existing cancer control knowledge into practice among all segments of the population. (*CA Cancer J Clin* 2004;54:8–29.) © American Cancer Society, 2004.

INTRODUCTION

Cancer is a major public health burden in the United States and in other developed countries. Currently, one in four deaths in the United States is due to cancer. In this article, we provide an overview of the most frequently used cancer statistics, including frequency, incidence, mortality, and survival rates for the year 2004.

MATERIALS AND METHODS

Data Sources

We obtained mortality data from 1930 to 2001 from the National Center for Health Statistics.¹ We obtained incidence data (1975 to 2000), five-year relative survival rates, and data on lifetime probability of developing

TABLE 1 Estimated New Cancer Cases and Deaths by Sex, US, 2004*

	Estimated New Cases			Estimated Deaths		
	Both Sexes	Male	Female	Both sexes	Male	Female
All Sites	1,368,030	699,560	668,470	563,700	290,890	272,810
Oral cavity & pharynx	28,260	18,550	9,710	7,230	4,830	2,400
Tongue	7,320	4,860	2,460	1,700	1,100	600
Mouth	10,080	5,410	4,670	1,890	1,070	820
Pharynx	8,250	6,330	1,920	2,070	1,460	610
Other oral cavity	2,610	1,950	660	1,570	1,200	370
Digestive system	255,640	135,410	120,230	134,840	73,240	61,600
Esophagus	14,250	10,860	3,390	13,300	10,250	3,050
Stomach	22,710	13,640	9,070	11,780	6,900	4,880
Small intestine	5,260	2,750	2,510	1,130	610	520
Colon†	106,370	50,400	55,970	56,730	28,320	28,410
Rectum	40,570	23,220	17,350			
Anus, anal canal, & anorectum	4,010	1,890	2,120	580	210	370
Liver & intrahepatic bile duct	18,920	12,580	6,340	14,270	9,450	4,820
Gallbladder & other biliary	6,950	2,960	3,990	3,540	1,290	2,250
Pancreas	31,860	15,740	16,120	31,270	15,440	15,830
Other digestive organs	4,740	1,370	3,370	2,240	770	1,470
Respiratory system	186,550	102,730	83,820	165,130	95,460	69,670
Larynx	10,270	8,060	2,210	3,830	3,010	820
Lung & bronchus	173,770	93,110	80,660	160,440	91,930	68,510
Other respiratory organs	2,510	1,560	950	860	520	340
Bones & joints	2,440	1,230	1,210	1,300	720	580
Soft tissue (including heart)	8,680	4,760	3,920	3,660	2,020	1,640
Skin (excluding basal & squamous)	59,350	31,640	27,710	10,250	6,590	3,660
Melanoma-skin	55,100	29,900	25,200	7,910	5,050	2,860
Other nonepithelial skin	4,910	2,400	2,510	2,340	1,540	800
Breast	217,440	1,450	215,990	40,580	470	40,110
Genital system	323,210	240,660	82,550	59,250	30,530	28,720
Uterine cervix	10,520		10,520	3,900		3,900
Uterine corpus	40,320		40,320	7,090		7,090
Ovary	25,580		25,580	16,090		16,090
Vulva	3,970		3,970	850		850
Vagina & other genital, female	2,160		2,160	790		790
Prostate	230,110	230,110		29,900	29,900	
Testis	8,980	8,980		360	360	
Penis & other genital, male	1,570	1,570		270	270	
Urinary system	98,400	68,290	30,110	25,880	17,060	8,820
Urinary bladder	60,240	44,640	15,600	12,710	8,780	3,930
Kidney & renal pelvis	35,710	22,080	13,630	12,480	7,870	4,610
Ureter & other urinary organs	2,450	1,570	880	690	410	280
Eye & orbit	2,090	1,130	960	180	110	70
Brain & other nervous system	18,400	10,540	7,860	12,690	7,200	5,490
Endocrine system	25,520	6,950	18,570	2,440	1,140	1,300
Thyroid	23,600	5,960	17,640	1,460	620	840
Other endocrine	1,920	990	930	980	520	460
Lymphoma	62,250	33,180	29,070	20,730	11,090	9,640
Hodgkin disease	7,880	4,330	3,550	1,320	700	620
Non-Hodgkin lymphoma	54,370	28,850	25,520	19,410	10,390	9,020
Multiple myeloma	15,270	8,090	7,180	11,070	5,430	5,640
Leukemia	33,440	19,020	14,420	23,300	12,990	10,310
Acute lymphocytic leukemia	3,830	2,110	1,720	1,450	820	630
Chronic lymphocytic leukemia	8,190	5,050	3,140	4,800	2,730	2,070
Acute myeloid leukemia	11,920	6,280	5,640	8,870	4,810	4,060
Chronic myeloid leukemia	4,600	2,700	1,900	1,570	940	630
Other leukemia‡	4,900	2,880	2,020	6,610	3,690	2,920
Other & unspecified primary sites†	31,090	15,930	15,160	45,170	22,010	23,160

*Rounded to the nearest 10; Excludes basal and squamous cell skin cancers and in situ carcinomas except urinary bladder.

Carcinoma in situ of the breast accounts for about 59,390 new cases annually, and in situ melanoma accounts for about 40,780 new cases annually.

† Estimated deaths for colon and rectum cancers are combined.

‡ More deaths than cases suggests lack of specificity in recording underlying causes of death on death certificates.

Source: Estimates of new cases are based on incidence rates from 1979 to 2000, National Cancer Institute's Surveillance, Epidemiology and End Results program, nine oldest registries. Estimates of deaths are based on data from US Mortality Public Use Data Tapes, 1969 to 2001, National Center for Health Statistics, Center for Disease Control and Prevention, 2003.

American Cancer Society, Surveillance Research, 2004.

TABLE 2 Estimated New Cancer Cases for Selected Cancers by State, US, 2004*

State	All Cases	Female Breast	Uterine Cervix	Colon & Rectum	Uterine Corpus	Leukemia	Lung & Bronchus	Melanoma of the Skin	Non-Hodgkin Lymphoma	Prostate	Urinary Bladder
AL	24,270	3,980	190	2,330	680	530	3,350	840	840	4,850	810
AK	1,890	270	†	210	60	†	240	70	80	230	90
AZ	23,560	3,980	190	2,490	510	590	2,760	1,180	950	3,920	1,140
AR	14,800	2,050	160	1,630	340	370	2,230	560	640	2,150	570
CA	134,300	21,860	1,210	13,880	3,920	3,240	15,650	5,020	5,550	23,160	5,730
CO	15,510	2,580	110	1,610	400	440	1,740	910	810	2,540	620
CT	17,010	2,850	80	1,710	450	400	2,000	700	760	3,310	660
DE	4,390	700	†	410	110	110	550	210	200	690	†
DC	2,860	590	†	340	170	†	300	70	60	620	90
FL	97,290	13,350	730	9,950	2,450	2,500	13,390	4,250	2,690	17,090	4,550
GA	35,430	6,080	350	3,420	970	790	5,050	1,460	1,320	5,700	1,520
HI	5,070	750	†	520	170	110	570	140	250	1,000	190
ID	5,460	920	†	540	170	140	660	280	250	1,080	330
IL	60,280	9,640	490	6,680	2,050	1,550	7,320	2,020	2,270	9,930	2,610
IN	32,160	4,790	130	3,520	910	790	4,490	1,320	1,430	5,390	1,230
IA	15,940	2,320	110	1,840	510	460	1,820	560	640	3,160	620
KS	12,940	1,880	80	1,480	400	340	1,690	630	640	2,690	660
KY	22,720	3,340	190	2,310	510	470	3,660	1,040	980	2,620	850
LA	23,540	3,930	190	2,560	510	550	3,160	700	980	3,690	760
ME	7,520	920	†	800	230	140	950	280	250	1,150	470
MD	25,310	4,090	220	2,820	740	650	3,180	980	1,040	4,080	1,140
MA	33,050	5,170	130	3,520	970	760	4,050	1,460	1,150	5,700	1,800
MI	48,220	7,270	350	4,920	1,420	1,210	6,160	1,670	2,040	8,540	2,370
MN	22,720	3,610	110	2,200	680	630	2,580	980	1,290	4,230	1,040
MS	15,120	2,480	110	1,610	280	300	2,230	420	390	3,390	470
MO	30,290	4,680	240	3,240	850	780	4,090	1,320	1,400	3,460	1,140
MT	5,000	590	†	470	110	140	650	210	200	1,080	330
NE	8,280	1,290	†	1,010	280	230	1,040	350	360	1,460	330
NV	10,990	1,620	80	1,240	170	260	1,570	490	420	2,000	520
NH	6,290	920	30	670	170	140	800	280	140	1,000	380
NJ	43,830	7,970	380	4,770	1,760	1,030	5,110	1,810	1,820	7,930	2,040
NM	7,550	1,020	†	830	230	170	750	280	310	1,690	330
NY	88,190	15,190	840	9,890	3,180	2,110	10,020	3,060	2,770	14,470	4,410
NC	40,240	5,870	320	4,120	1,190	930	5,710	1,740	1,480	7,160	1,470
ND	3,250	540	†	360	60	100	360	70	140	540	190
OH	59,410	10,070	320	6,760	1,880	1,450	7,720	2,300	2,410	8,620	2,940
OK	18,540	2,910	130	2,070	400	440	2,570	910	760	2,620	760
OR	17,280	2,750	110	1,790	450	400	2,140	910	920	2,920	900
PA	72,590	11,200	400	8,570	2,500	1,620	8,560	2,720	3,030	12,010	3,510
RI	5,950	860	†	650	110	130	760	280	280	1,000	330
SC	21,500	3,280	160	2,280	510	490	2,950	700	870	4,770	810
SD	4,000	540	†	490	110	110	450	210	220	920	140
TN	30,850	4,310	300	3,470	740	730	4,680	1,250	1,400	4,540	1,090
TX	84,530	12,980	1,030	9,220	2,390	2,140	10,470	3,550	2,970	13,540	3,270
UT	6,360	1,080	†	670	230	220	480	420	390	1,080	280
VT	3,150	590	†	340	110	70	400	140	170	460	140
VA	31,190	6,350	220	3,550	1,080	760	4,050	1,390	1,230	5,080	1,330
WA	27,380	4,040	130	2,720	910	720	3,520	1,320	1,290	4,850	1,330
WV	11,430	1,620	110	1,270	340	270	1,780	420	500	1,540	570
WI	26,160	4,040	110	2,900	850	750	3,050	1,110	1,290	3,850	1,280
WY	2,430	270	†	280	60	60	280	140	80	620	90
US	1,368,030	215,990	10,520	146,940	40,320	33,440	173,770	55,100	54,370	230,110	60,240

* Rounded to the nearest 10. Excludes basal and squamous cell skin cancers and *in situ* carcinomas except urinary bladder.

† Estimate is 50 or fewer cases. Note: These estimates are offered as a rough guide and should be interpreted with caution. They are calculated according to the distribution of estimated cancer deaths in 2004 by state. State estimates may not add to US total due to rounding.

American Cancer Society, Surveillance Research, 2004.

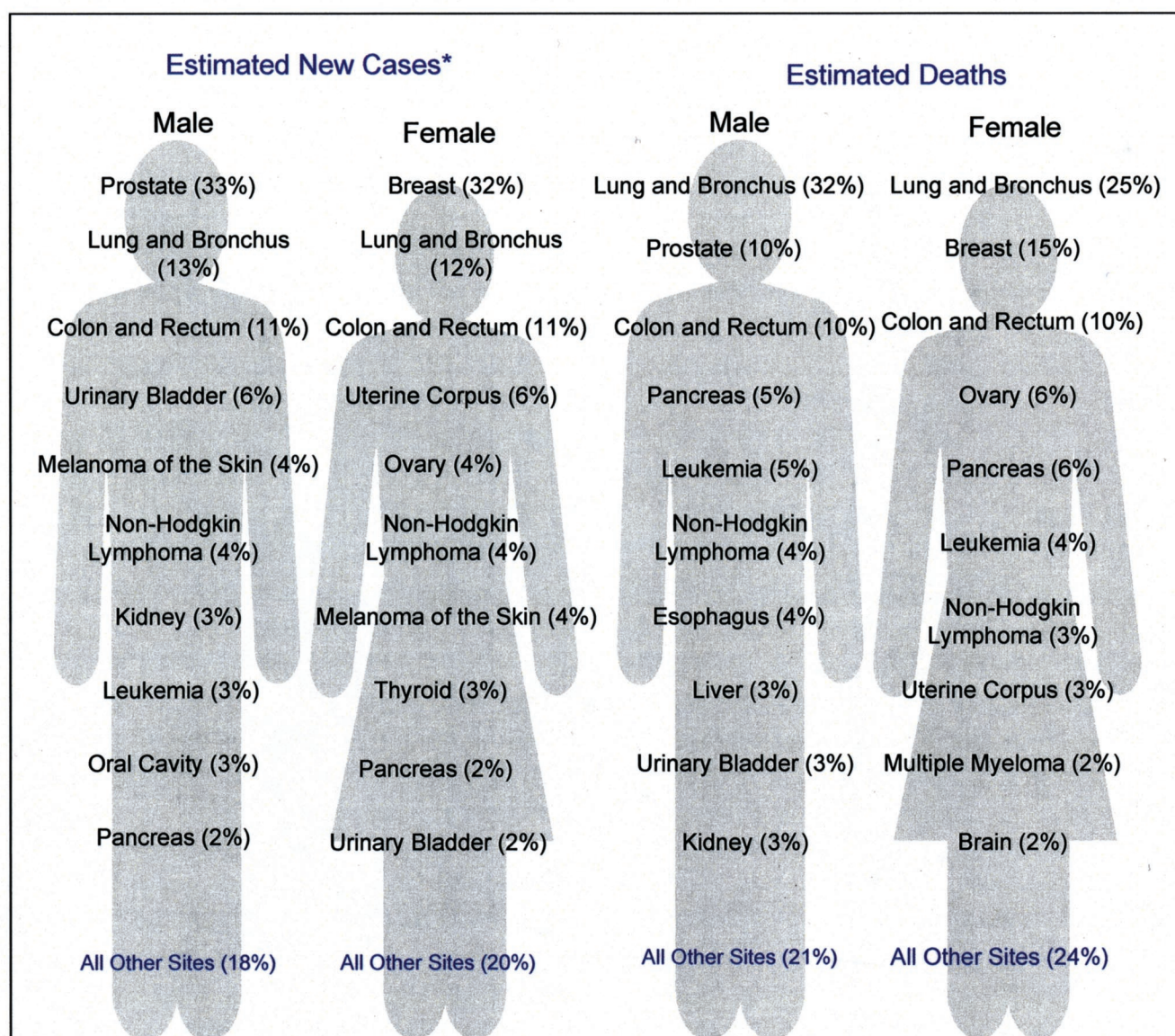


FIGURE 1 Ten Leading Cancer Types for the Estimated New Cancer Cases and Deaths, by Sex, US, 2004.*

*Excludes basal and squamous cell skin cancers and in situ carcinomas except urinary bladder.

Note: Percentages may not total 100 due to rounding.

American Cancer Society, Surveillance Research, 2004.

cancer from the Surveillance Epidemiology and End Results (SEER) program of the National Cancer Institute, covering approximately 10% to 14% of the US population.²⁻⁵ We obtained population data from the US Census Bureau.⁶ Causes of death were coded and classified according to the International Classification of Diseases (ICD-8, ICD-9, and ICD-10).⁷⁻⁹ Cancer cases were classified according to the second edition of the Inter-

national Classification of Diseases for Oncology.¹⁰

Estimated New Cancer Cases

The precise number of cancer cases diagnosed each year in the nation and in every state is unknown because complete cancer registration has not yet been achieved in many states. Consequently, for the national

TABLE 3 Age-standardized Death Rates and Estimated Cancer Deaths from Selected Cancers by State, US, 2004*

STATE	Death Rate† per 100,000	All Sites	Brain/Nervous System	Female Breast	Colon & Rectum	Leukemia	Liver	Lung & Bronchus	Non-Hodgkin Lymphoma	Ovary	Pancreas	Prostate
AL	215.9	10,000	200	740	900	370	260	3,090	300	320	530	630
AK	204.8	780	†	†	80	†	†	220	†	†	†	†
AZ	178.9	9,710	240	740	960	410	260	2,550	340	290	560	510
AR	213.2	6,100	160	380	630	260	200	2,060	230	150	280	280
CA	186.2	55,340	1,440	4,060	5,360	2,260	1,880	14,450	1,980	1,730	3,020	3,010
CO	174.4	6,390	180	480	620	310	160	1,610	290	210	380	330
CT	194.3	7,010	150	530	660	280	170	1,850	270	200	380	430
DE	220.6	1,810	†	130	160	80	†	510	70	†	110	90
DC	244.5	1,180	†	110	130	†	†	280	†	†	60	80
FL	193.6	40,090	980	2,480	3,840	1,740	1,030	12,360	960	1,120	2,270	2,220
GA	205.7	14,600	260	1,130	1,320	550	300	4,660	470	430	750	740
HI	159.9	2,090	†	140	200	80	100	530	90	†	150	130
ID	182.7	2,250	70	170	210	100	†	610	90	80	120	140
IL	211.1	24,840	490	1,790	2,580	1,080	650	6,760	810	660	1,400	1,290
IN	215.4	13,250	280	890	1,360	550	280	4,150	510	380	670	700
IA	191.7	6,570	160	430	710	320	110	1,680	230	210	380	410
KS	190.3	5,330	120	350	570	240	100	1,560	230	160	300	350
KY	228.9	9,360	160	620	890	330	180	3,380	350	230	410	340
LA	232.8	9,700	190	730	990	380	280	2,920	350	230	520	480
ME	214.9	3,100	80	170	310	100	60	880	90	100	170	150
MD	214.9	10,430	210	760	1,090	450	240	2,940	370	300	590	530
MA	209.0	13,620	280	960	1,360	530	340	3,740	410	360	830	740
MI	206.6	19,870	450	1,350	1,900	840	500	5,690	730	580	1,120	1,110
MN	190.0	9,360	250	670	850	440	190	2,380	460	270	540	550
MS	223.7	6,230	160	460	620	210	190	2,060	140	160	320	440
MO	208.8	12,480	270	870	1,250	540	270	3,780	500	350	660	450
MT	194.6	2,060	†	110	180	100	†	600	70	†	100	140
NE	185.7	3,410	90	240	390	160	†	960	130	90	180	190
NV	211.8	4,530	80	300	480	180	110	1,450	150	110	220	260
NH	209.8	2,590	70	170	260	100	60	740	†	60	140	130
NJ	211.7	18,060	320	1,480	1,840	720	480	4,720	650	540	1,040	1,030
NM	174.9	3,110	70	190	320	120	110	690	110	90	170	220
NY	198.1	36,340	690	2,820	3,820	1,470	890	9,250	990	1,080	2,270	1,880
NC	207.0	16,580	320	1,090	1,590	650	350	5,270	530	450	900	930
ND	186.2	1,340	40	100	140	70	†	330	†	†	90	70
OH	215.5	24,480	520	1,870	2,610	1,010	520	7,130	860	660	1,290	1,120
OK	207.0	7,640	160	540	800	310	160	2,370	270	170	360	340
OR	200.4	7,120	160	510	690	280	150	1,980	330	230	400	380
PA	209.9	29,910	570	2,080	3,310	1,130	690	7,900	1,080	910	1,650	1,560
RI	212.7	2,450	†	160	250	90	60	700	100	60	160	130
SC	211.6	8,860	200	610	880	340	190	2,720	310	170	500	620
SD	192.1	1,650	†	100	190	80	†	420	80	60	100	120
TN	217.8	12,710	300	800	1,340	510	270	4,320	500	340	660	590
TX	199.8	34,830	940	2,410	3,560	1,490	1,120	9,670	1,060	960	1,930	1,760
UT	152.1	2,620	80	200	260	150	60	440	140	90	150	140
VT	204.0	1,300	†	110	130	50	†	370	60	†	70	60
VA	209.8	12,850	290	1,180	1,370	530	320	3,740	440	400	750	660
WA	196.7	11,280	340	750	1,050	500	300	3,250	460	390	700	630
WV	224.7	4,710	90	300	490	190	100	1,640	180	140	190	200
WI	195.7	10,780	260	750	1,120	520	250	2,820	460	260	630	500
WY	193.0	1,000	†	†	110	†	†	260	†	†	†	80
US	202.3	563,700	12,690	40,110	56,730	23,300	14,270	160,440	19,410	16,090	31,270	29,900

* Rounded to the nearest 10; excludes in situ carcinomas except urinary bladder.

† Estimate is 50 or fewer deaths.

‡ Average annual rates for 1996 to 2000 and age adjusted to 2000 US standard population.

Note: State estimates may not add to US total due to rounding.

Source: US Mortality Public Use Data Tapes, 1969 to 2001, National Center for Health Statistics, Centers for Disease Control and Prevention, 2003.

American Cancer Society, Surveillance Research, 2004.

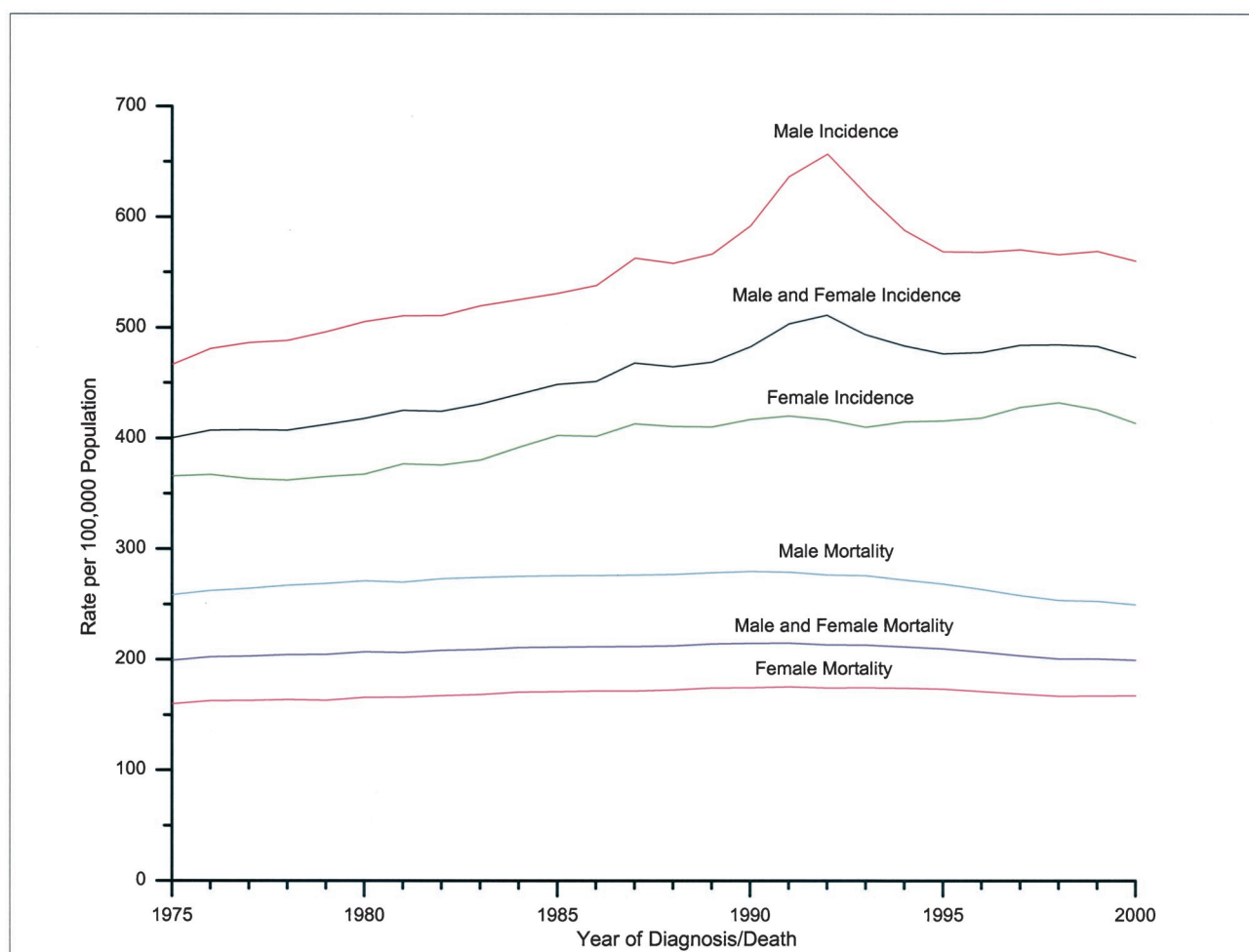


FIGURE 2 Annual Age-adjusted Cancer Incidence and Death Rates* All Sites, by Sex, US, 1975 to 2000.

*Rates are age-adjusted to the 2000 US standard population. Source: Incidence data from Surveillance, Epidemiology, and End Results (SEER) program, nine oldest registries, 1973 to 2000, Division of Cancer Control and Population Sciences, National Cancer Institute, 2003.

Mortality data from US Mortality Public Use Data Tapes, 1960 to 2000, National Center for Health Statistics, Centers for Disease Control and Prevention.

estimate, we first estimate the number of new cancer cases occurring annually in the United States from 1979 through 2000 by using age-specific cancer incidence rates collected by the SEER program² coupled with population data reported by the US Census Bureau.⁶ We then forecast the number of cancer cases expected to be diagnosed in the United States in the year 2004 using an autoregressive quadratic time trend model fitted to the annual cancer case estimates.¹¹ For estimates of new cancer cases in individual states, we relied on state cancer death statistical data and assumed that the ratio of estimated cancer deaths to cases in each state was equal to that in the United States.

Estimated Cancer Deaths

For this report, we used a new method for estimating the number of cancer deaths expected to occur in the United States and in each state in the year 2004. The method is described by Tiwari et al. in an accompanying article in this issue of *CA*.¹² We based our projections on underlying causes of death recorded on death certificates as reported to the National Center for Health Statistics.¹ The numbers of cancer deaths recorded annually from 1969 to 2001 in the United States and in each state were fitted using a state-space model to forecast the number of cancer deaths expected to occur in 2004.

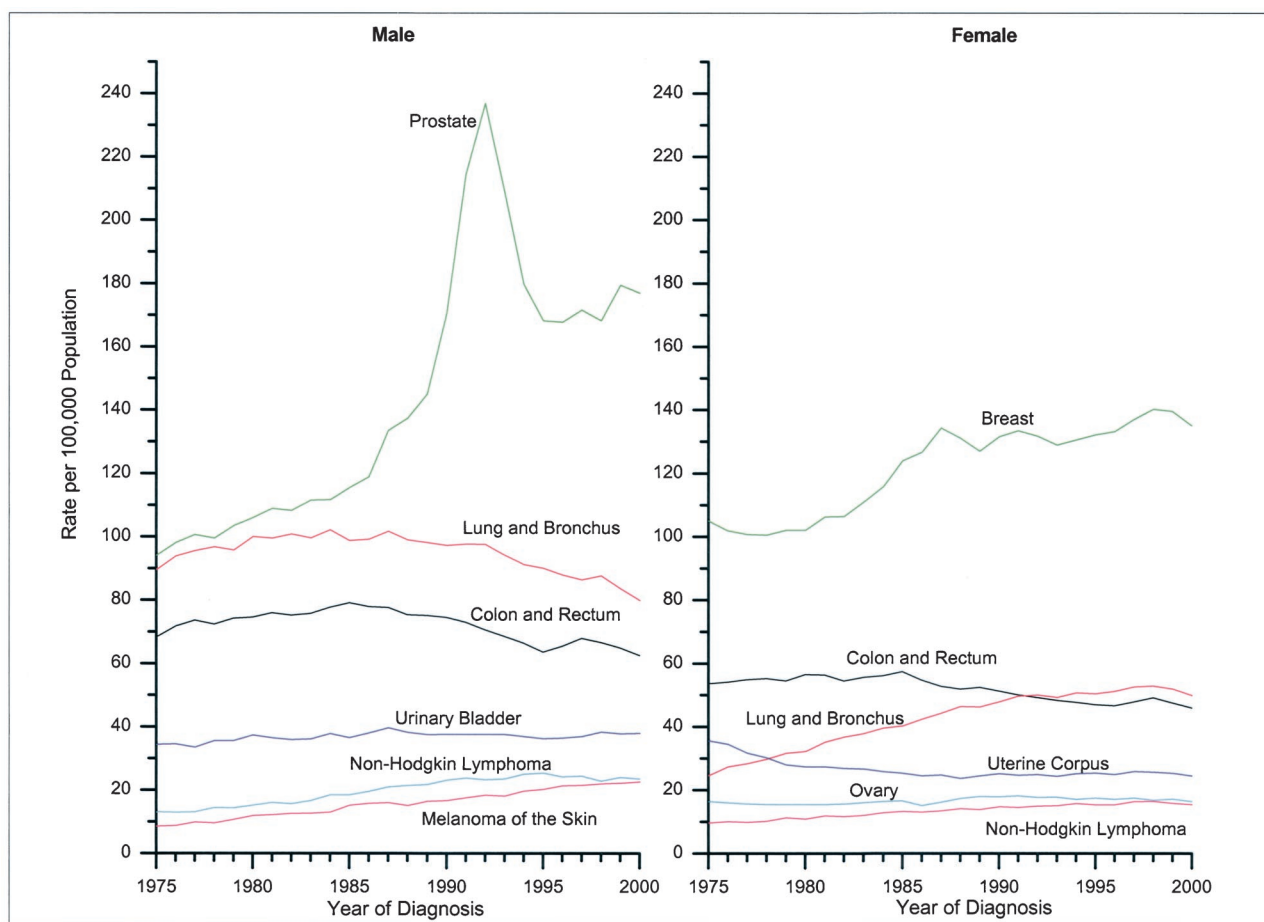


FIGURE 3 Annual Age-adjusted Cancer Incidence Rates* Among Males and Females for Selected Cancer Types, US, 1975 to 2000.

*Rates are age-adjusted to the 2000 US standard population.

Source: Surveillance, Epidemiology, and End Results (SEER) program, nine oldest registries, 1973 to 2000, Division of Cancer Control and Population Sciences, National Cancer Institute, 2003.

Other Statistics

We provide mortality statistics for the leading causes of death and deaths due to cancer in the year 2001. Causes of death for 2001 were coded and classified according to ICD-10.⁷ This report also provides updated statistics on trends in cancer incidence and mortality rates, probability of developing cancer, and five-year relative survival rates for selected cancer sites based on data from 1973 through 2000.³ All age-adjusted incidence and mortality rates are standardized to the 2000 US standard population and expressed per 100,000 population.

The long-term incidence trends (1975 to 2000) presented in the table are adjusted for delays in reporting. Delayed reporting affects the most recent one to three years of incidence data (in this case, 1998 to 2000) as certain types

of cancer continue to be reported. Cancers that are frequently reported late are those diagnosed in outpatient settings, such as melanoma and prostate cancers. The National Cancer Institute has developed a method to account for expected reporting delays for all cancer sites combined and for several specific cancer sites when long-term incidence trends are analyzed.¹³ Delay-adjusted trends provide a more accurate assessment of trends in the most recent years for which data are available.

SELECTED FINDINGS

Expected Numbers of New Cancer Cases

Table 1 presents estimated numbers of new cancer cases expected in the United States in 2004 by sex. The estimate of approximately 1.4

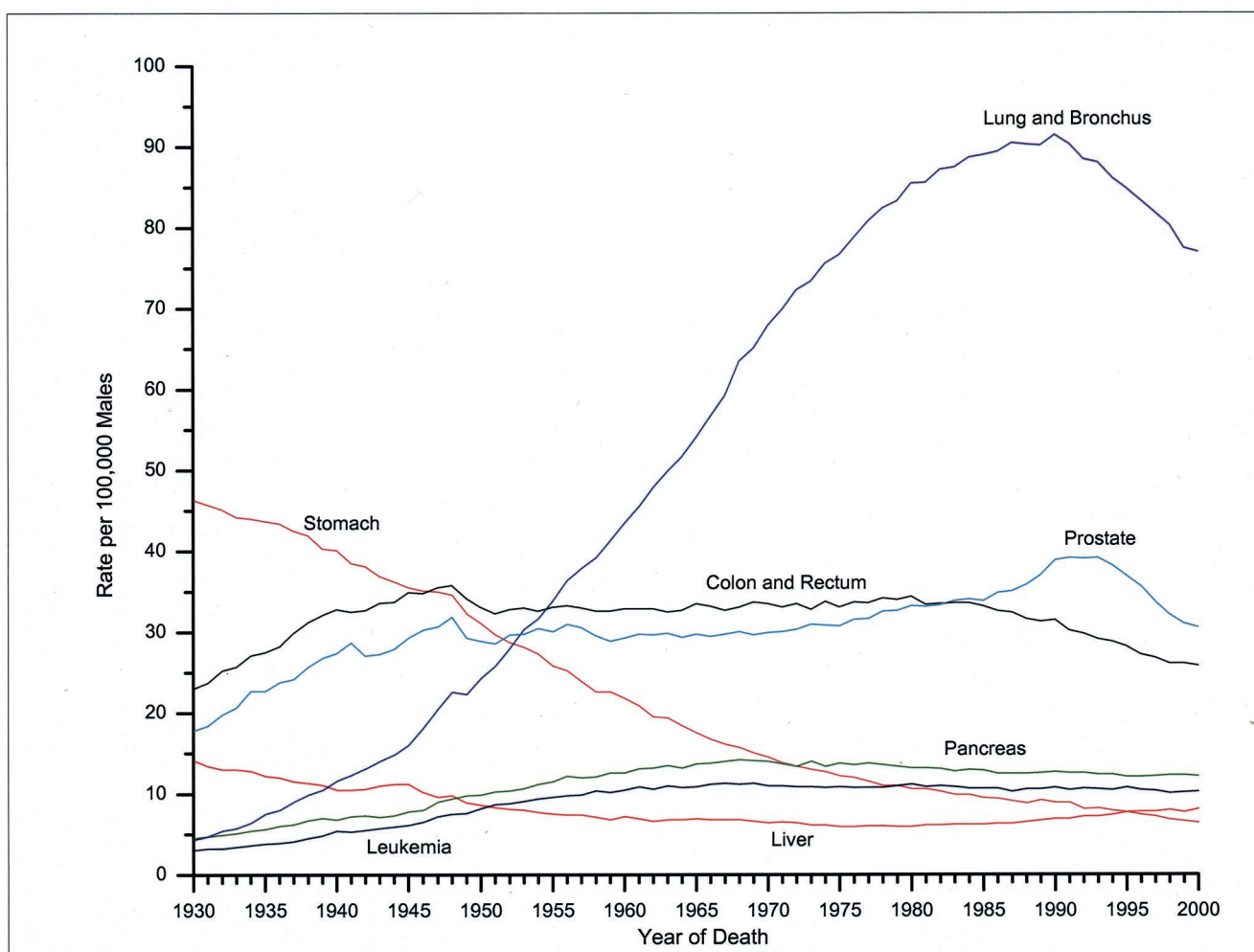


FIGURE 4 Annual Age-adjusted Cancer Death Rates* Among Males for Selected Cancer Types, US, 1930 to 2000.

*Rates are age-adjusted to the 2000 US standard population.

Note: Due to changes in ICD coding, numerator information has changed over time. Rates for cancers of the lung and bronchus and colon and rectum are affected by these changes.

Source: US Mortality Public Use Data Tapes, 1960 to 2000, US Mortality Volumes, 1930 to 1959, National Center for Health Statistics, Centers for Disease Control and Prevention.

million new cases of invasive cancer does not include carcinoma in situ of any site except the urinary bladder, nor does it include basal and squamous cell cancers of the skin. More than one million cases of basal and squamous cell skin cancer, approximately 59,390 cases of breast carcinoma in situ, and 40,780 cases of in situ melanoma are expected to be newly diagnosed in 2004. Table 2 shows the estimated numbers of new cancer cases for each state and selected cancer sites.

Figure 1 lists the most common cancers expected to occur in men and women in 2004. Among men, cancers of the prostate,

lung and bronchus, and colon and rectum account for more than 55% of all newly diagnosed cancers. Prostate cancer alone accounts for approximately 33% (230,110) of incident cases in men. Based on cases diagnosed between 1992 and 1999, approximately 86% of these estimated new cases of prostate cancer are expected to be diagnosed at local or regional stages, for which the five-year relative survival rate equals 100%.

The three most commonly diagnosed cancers among women in 2004 will be those of the breast, lung and bronchus, and colon and rectum, accounting for approximately

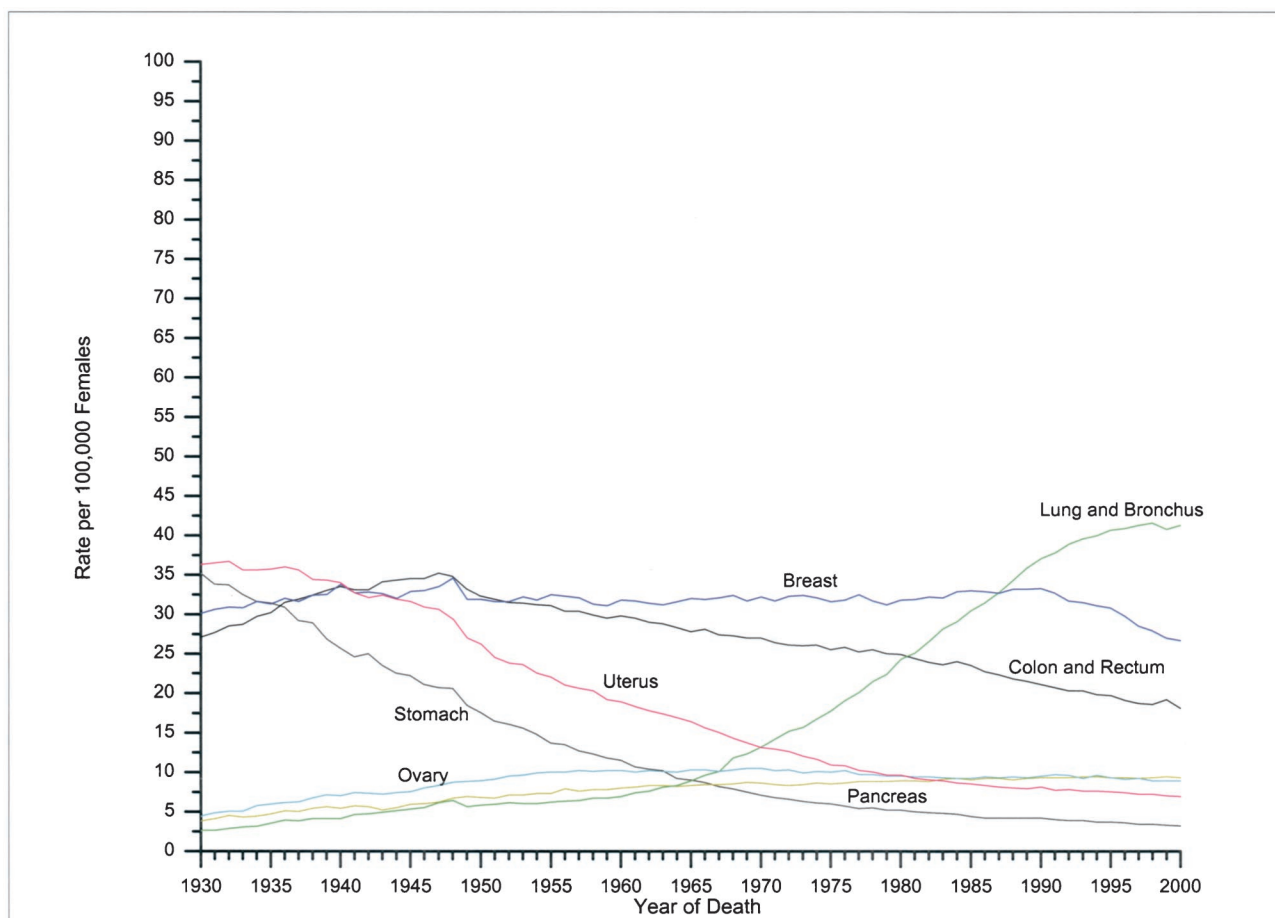


FIGURE 5 Annual Age-adjusted Cancer Death Rates* Among Female for Selected Cancer Types, US, 1930 to 2000.

*Rates are age-adjusted to the 2000 US standard population.

Note: Due to ICD coding, numerator information has changed over time. Rates for cancers of the uterus, ovary, lung and bronchus, and colon and rectum are affected by these changes. Uterus cancers are for uterine cervix and uterine corpus combined.

Source: US Mortality Public Use Data Tapes, 1960 to 2000, US Mortality Volumes 1930 to 1959, National Center for Health Statistics, Centers for Disease Control and Prevention.

55% of estimated cancer cases in women. Breast cancer alone is expected to account for 32% (215,990) of all new cancer cases among women.

Expected Numbers of New Cancer Deaths

Table 1 also shows the expected numbers of cancer deaths in 2004 for men, women, and both sexes combined. It is estimated that approximately 563,700 Americans will die of cancer, corresponding to over 1,500 deaths per day. Cancers of the lung and bronchus, prostate, and colon and rectum in men, and cancers of the lung and bronchus, breast, and colon and rectum in women continue to be the most common causes of cancer death. These four

cancers account for one half of the total number of cancer deaths among men and women (Figure 1). Lung cancer surpassed breast cancer as the leading cause of cancer death in women in 1987. Lung cancer is expected to account for 25% of all female cancer deaths in 2004. Table 3 provides the estimated number of cancer deaths in 2004 by state for selected cancer sites.

Trends in Cancer Incidence and Mortality Rates

Figures 2 to 5 depict long-term trends in cancer incidence and mortality rates for all cancers combined and for selected cancer sites by sex. Table 4 shows incidence and mortality patterns for all cancer sites and for the four common cancer sites based on joinpoint anal-

TABLE 4 Trends in Cancer Incidence and Death Rates for Selected Cancers by Sex, All Races, 1975 to 2000

	Line Segment 1		Line Segment 2		Line Segment 3		Line Segment 4	
	Year	APC	Year	APC	Year	APC	Year	APC
All Cancers								
Incidence								
Male and female	1978-83	0.9*	1983-92	1.9*	1992-95	-1.8	1995-00	0.6
Male	1975-89	1.4*	1989-92	5.2*	1992-95	-4.8*	1995-00	0.5
Female	1975-79	-0.3	1979-87	1.5*	1987-00	0.4*		
Death								
Male and female	1975-90	0.5*	1990-94	-0.4	1994-98	-1.4*	1998-00	-0.3
Male	1975-80	0.9*	1980-92	0.2*	1992-00	-1.5*		
Female	1975-90	0.6*	1990-95	-0.2	1995-98	-1.3	1998-00	0.1
Lung & bronchus								
Incidence								
Male and female	1975-82	2.5*	1982-91	1.0*	1991-00	-0.7*		
Male	1975-82	1.4*	1982-91	-0.4	1991-00	-1.8*		
Female	1975-82	5.6*	1982-91	3.4*	1991-00	0.7*		
Death								
Male and female	1975-82	2.7*	1982-91	1.6*	1991-00	-0.7*		
Male	1975-82	1.8*	1982-91	0.4*	1991-00	-1.8*		
Female	1975-83	5.9*	1983-92	3.8*	1992-00	0.6*		
Colon & rectum								
Incidence								
Male and female	1975-85	0.8*	1985-95	-1.8*	1995-98	1.2	1998-00	-2.3
Male	1975-86	1.1*	1986-95	-2.0*	1995-00	-0.2		
Female	1975-85	0.3	1985-94	-1.8*	1994-00	0.2		
Death								
Male and female	1975-84	-0.5*	1984-00	-1.7*				
Male	1975-79	0.6	1979-87	-0.6*	1987-00	-1.9*		
Female	1975-84	-1.0*	1984-00	-1.8*				
Female breast								
Incidence	1975-80	-0.6	1980-86	4.1*	1986-00	0.6*		
Death	1975-90	0.4*	1990-00	-2.3*				
Prostate								
Incidence	1975-88	2.6*	1988-92	16.6*	1992-95	-11.6*	1995-00	2.3*
Death	1975-87	0.9*	1987-91	3.1*	1991-94	-0.6	1994-00	-4.0*

*The APC is significantly different from zero ($p < 0.05$).

APC = Annual percent change based on rates age-adjusted to the 2000 US standard population.

Note: Trends were analyzed by Joinpoint Regression Program, version 2.7, with a maximum of three jointpoints, i.e., four line segments.

Incidence is adjusted for delay in reporting.

Source: Ries LAG, Eisner MP, Kosary CL, et al.³

ysis. Trends in incidence were adjusted for delayed reporting in the table, but not in the figures (Figures 2 and 3). Cancer death rates for all cancer sites combined decreased by 1.5% per year from 1992 to 2000 in males, but they stabilized from 1998 to 2000 in females after declining by 1.3% from 1995 to 1998 (Table 4). The leveling of rates among females is in part due to the implementation of ICD-10

rules for coding and selecting the underlying cause of death. Approximately 0.7% more deaths were classified as cancer using ICD-10 rules than using ICD-9 rules,¹⁴ creating discontinuity in the most recent cancer mortality rate trend.^{15,16} Delay-adjusted cancer incidence rates stabilized in men from 1995 to 2000 and increased by 0.4% per year from 1987 to 2000 in women (Table 4).

TABLE 5 Fifteen Leading Causes of Death, US, 2001

Rank	Cause of Death	Number of Deaths	Percent (%) of Total Deaths	Death Rate*
	All Causes	2,416,425		854.2
1	Heart Diseases	700,142	29.0	246.6
2	Cancer	553,768	22.9	196.7
3	Cerebrovascular Diseases	163,538	6.8	57.5
4	Chronic Lower Respiratory Diseases	123,013	5.1	43.5
5	Accidents (Unintentional Injuries)	101,537	4.2	36.2
6	Diabetes Mellitus	71,372	3.0	25.3
7	Influenza & Pneumonia	62,034	2.6	21.8
8	Alzheimer Disease	53,852	2.2	18.8
9	Nephritis, Nephrotic Syndrome, & Nephrosis	39,480	1.6	13.9
10	Septicemia	32,238	1.3	11.4
11	Intentional Self-harm (Suicide)	30,618	1.3	11.0
12	Chronic Liver Disease & Cirrhosis	27,035	1.1	9.7
13	Hypertension and Hypertensive Renal Disease	19,250	0.8	6.8
14	Assault (Homicide)	17,386	0.7	6.3
15	Parkinsons Disease	16,544	0.7	5.8
	All Other & Ill-defined Causes	404,618	16.7	

*Rates are per 100,000 population and age-adjusted to the 2000 US standard population.

Note: Percentages may not total 100 due to rounding. Symptoms, Signs, and Abnormalities were excluded from the cause of death ranking order.

Source: US Mortality Public Use Data Tape, 2001, National Center for Health Statistics, Centers for Disease Control and Prevention, 2003.

Mortality rates have continued to decrease across all four major cancer sites in men and in women (Table 4). The incidence trends are mixed, however. Lung cancer incidence rates are declining in men but still increasing among women, although the rate of increase has slowed. Colorectal cancer incidence rates have stabilized since the mid 1990s in males and females. Female breast and prostate cancer incidence rates have continued to increase, although at a slower rate than in previous years, in part due to increased screening using mammography and prostate-specific antigen testing, respectively. The recent increase in female breast cancer incidence rates involves both small and large tumors,¹⁷ which may reflect increased use of hormone replacement therapy, an increased prevalence of obesity, or both.

Changes in the Recorded Number of Deaths Due to Cancer From 2000 to 2001

A total of 553,768 cancer deaths were recorded in the United States in 2001, the most recent year for which actual data are available. Nearly 700 more cancer deaths were recorded in 2001 than in 2000, reflecting growth and aging of the population. Cancer accounted for approximately 23% of all deaths, ranking second only to heart disease (Table 5). When cause of death is ranked within each age group, categorized in 20-year age intervals, cancer ranks as one of the five leading causes of death in each age group among both males and females. Cancer is the leading cause of death among women aged 40 to 79 years and among men aged 60 to 79 years (Table 6). Overall, among men and women combined, cancer is the leading cause of death

TABLE 6 Ten Leading Causes of Death by Age and Sex, US, 2001

	All Ages		Ages 1 to 19		Ages 20 to 39		Ages 40 to 59		Ages 60 to 79		Ages 80+	
	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female
	All Causes	All Causes	All Causes	All Causes	All Causes	All Causes	All Causes	All Causes	All Causes	All Causes	All Causes	All Causes
	1,183,421	1,233,004	16,833	8,924	66,681	30,594	205,086	127,281	492,049	407,204	386,964	646,819
1	Heart Diseases	Heart Diseases	Accidents (Unintentional Injuries)	Accidents (Unintentional Injuries)	Accidents (Unintentional Injuries)	Accidents (Unintentional Injuries)	Heart Diseases	Cancer	Cancer	Cancer	Heart Diseases	Heart Diseases
	339,095	361,047	7,570	3,626	20,704	6,489	51,788	49,065	157,504	129,877	134,089	234,243
2	Cancer	Cancer	Assault (Homicide)	Cancer	Intentional Self-harm (Suicide)	Cancer	Cancer	Heart Diseases	Heart Diseases	Heart Diseases	Cancer	Cancer
	287,075	266,693	2,058	951	8,781	5,664	51,427	21,082	146,916	102,404	72,129	81,100
3	Accidents (Unintentional Injury)	Cerebrovascular Disease	Intentional Self-harm (Suicide)	Assault (Homicide)	Assault (Homicide)	Heart Diseases	Accidents (Unintentional Injuries)	Accidents (Unintentional Injuries)	Chronic Lower Respiratory Diseases	Chronic Lower Respiratory Diseases	Cerebrovascular Diseases	Cerebrovascular Diseases
	66,060	100,361	1,559	573	7,619	2,701	18,725	7,134	31,853	30,330	30,560	67,556
4	Cerebrovascular Disease	Chronic Lower Respiratory Diseases	Cancer	Congenital Anomalies	Heart Diseases	Intentional Self-harm (Suicide)	Chronic Liver Disease & Cirrhosis	Cerebrovascular Diseases	Cerebrovascular Diseases	Cerebrovascular Diseases	Chronic Lower Respiratory Diseases	Alzheimer's Disease
	63,177	63,316	1,209	535	5,530	1,825	8,949	5,366	25,346	26,539	23,170	31,852
5	Chronic Lower Respiratory Diseases	Diabetes Mellitus	Congenital Anomalies	Heart Diseases	Cancer	Assault (Homicide)	Intentional Self-harm (Suicide)	Diabetes Mellitus	Diabetes Mellitus	Diabetes Mellitus	Influenza & Pneumonia	Chronic Lower Respiratory Diseases
	59,697	38,531	653	362	4,768	1,799	8,700	4,493	16,560	17,099	15,801	28,541
6	Diabetes Mellitus	Alzheimers Disease	Heart Disease	Intentional Self-harm (Suicide)	HIV Disease	HIV Disease	Cerebrovascular Diseases	Chronic Lower Respiratory Diseases	Accidents (Unintentional Injuries)	Nephritis, Nephrosis, & Nephrotic Syndrome	Alzheimer's Disease	Influenza & Pneumonia
	32,841	38,090	482	331	3,547	1,478	6,288	3,969	10,417	7,478	11,403	25,557
7	Influenza & Pneumonia	Accidents (Unintentional Injury)	Influenza & Pneumonia	Influenza & Pneumonia	Chronic liver Disease & cirrhosis	Cerebrovascular Disease	Diabetes Mellitus	Chronic Liver Disease & Cirrhosis	Influenza & Pneumonia	Influenza & pneumonia	Diabetes Mellitus	Diabetes Mellitus
	27,342	35,477	140	130	982	756	6,261	3,537	8,628	7,081	9,131	16,299
8	Intentional Self-harm (Suicide)	Influenza & Pneumonia	Chronic Lower Respiratory Diseases	In Situ/Benign/Unk. Neopl.	Diabetes Mellitus	Diabetes Mellitus	HIV Disease	Intentional Self-harm (Suicide)	Nephritis, Nephrotic Syndrome & Nephrosis	Accidents (Unintentional Injuries)	Nephritis, Nephrotic Syndrome & Nephrosis	Nephritis, Nephrotic Syndrome & Nephrosis
	24,668	34,692	133	108	835	598	6,232	2,686	7,863	6,889	8,525	11,135
9	Nephritis, Nephrotic Syndrome & Nephrosis	Nephritis, Nephrotic Syndrome & Nephrosis	Septicemia	Septicemia	Cerebrovascular Disease	Chronic Liver Disease & Cirrhosis	Chronic Lower Respiratory Diseases	Septicemia	Chronic Liver Disease & Cirrhosis	Septicemia	Accidents (Unintentional Injuries)	Accidents (Unintentional Injuries)
	18,852	20,628	130	104	813	533	4,145	1,837	6,353	6,393	7,999	10,888
10	Chronic Liver Disease & Cirrhosis	Septicemia	Cerebrovascular Disease	Cerebrovascular Disease	Congenital Anomalies	Events of Undetermined Intent	Assault	HIV Disease	Septicemia	Alzheimer Disease	Parkinsons Disease	Septicemia
	17,393	17,931	103	99	531	515	2732	1835	6,324	6,099	5,648	9,146

Note: Symptoms, Signs, and Abnormalities and Events of Undetermined Intent were excluded from the cause of death ranking order.

Source: US Mortality Public Use Data Tapes, 2001, National Center for Health Statistics, Centers for Disease Control and Prevention, 2003.

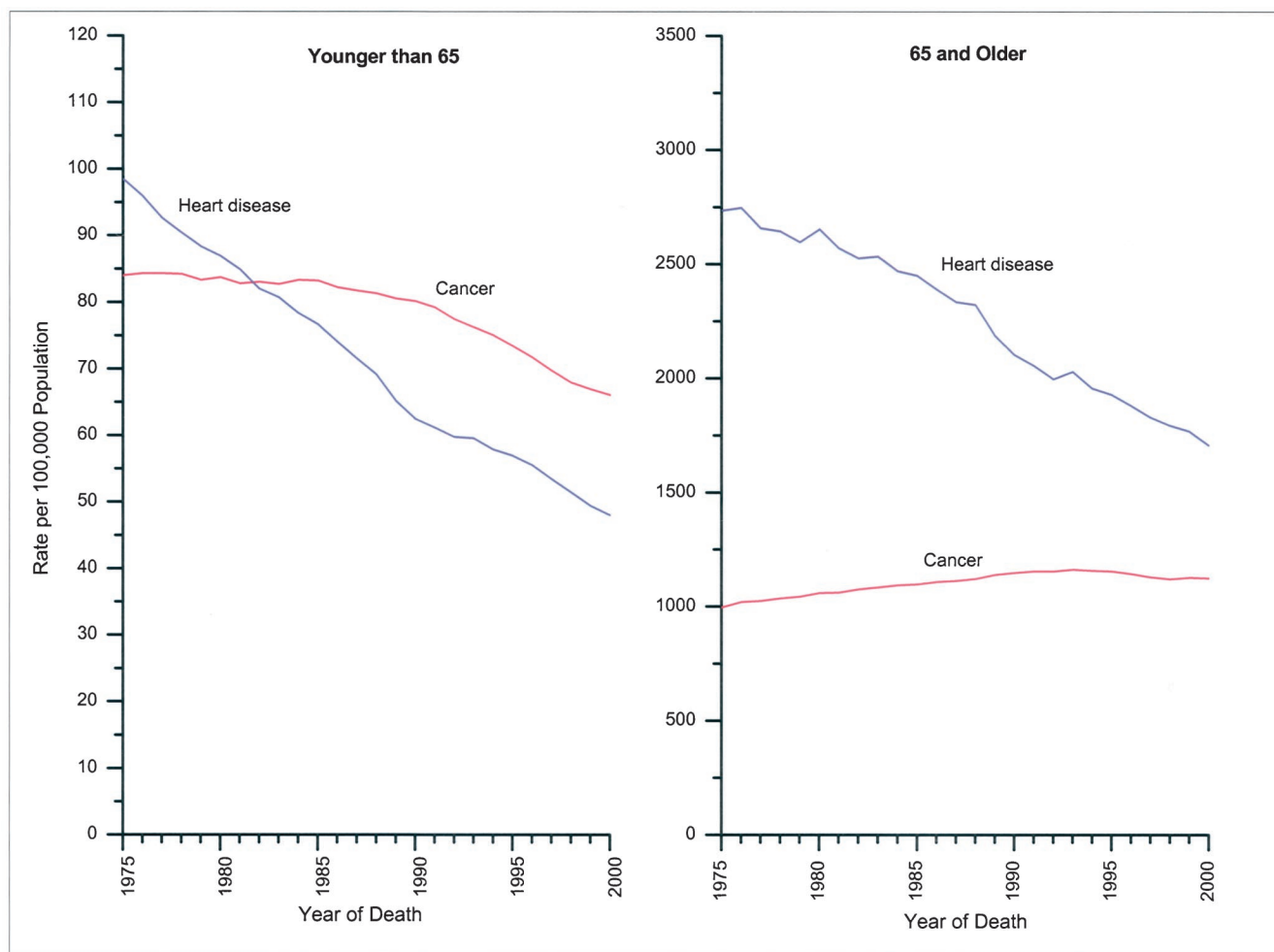


FIGURE 6 Death Rates* From Cancer and Heart Disease for Ages Younger than 65 and 65 and Older.

*Rates are age-adjusted to the 2000 US standard population.

Source: US Mortality Public Use Data Tapes, 1960 to 2000, National Center for Health Statistics, Centers for Disease Control and Prevention, 2003.

before age 65, followed by heart disease (Figure 6). A total of 163,546 persons younger than 65 years died of cancer in the United States in 2001, compared with 117,346 persons who died of heart disease.

Table 7 presents the leading site-specific causes of cancer death by age for males and females in 2001. Among men younger than 40 years, leukemia is the most common fatal cancer, whereas cancer of the lung and bronchus predominate in men aged 40 years and older. Colorectal cancer is the second most common fatal cancer among men aged 40 to 79 years, and prostate cancer is the second most common among men aged 80 and older. Among women, the leading causes of cancer death are leukemia among those younger than

20 years, breast cancer among those aged 20 to 59 years, and lung cancer in those 60 years and older.

From 2000 to 2001, the number of recorded cancer deaths increased by 993 in men, whereas it decreased by 316 in women. The total number of cancer deaths decreased slightly across all of the four major cancer sites, except for lung cancer in females. Five hundred ninety more women in the United States died of lung cancer in 2001 than in 2000 (Table 8).

Cancer Occurrence by Race and Ethnicity

Cancer incidence and mortality rates vary considerably among racial and ethnic groups (Table

TABLE 7 Reported Deaths for the Five Leading Cancer Sites by Age and Sex, US, 2001

All Ages	<20	20 to 39	40 to 59	60 to 79	≥ 80
Male					
All Sites 287,075	All Sites 1,240	All Sites 4,768	All Sites 51,427	All Sites 157,504	All Sites 72,129
Lung & Bronchus 90,367	Leukemia 410	Leukemia 654	Lung & Bronchus 15,886	Lung & Bronchus 56,981	Lung & Bronchus 17,088
Prostate 30,719	Brain & ONS 268	Brain & ONS 573	Colon & Rectum 5,078	Colon & Rectum 14,903	Prostate 15,706
Colon & Rectum 28,229	Bones & Joints 114	Colon & Rectum 454	Pancreas 2,958	Prostate 13,922	Colon & Rectum 7,787
Pancreas 14,467	Endocrine System 96	Non-Hodgkin Lymphoma 425	Liver 2,472	Pancreas 8,180	Urinary Bladder 3,313
Leukemia 11,894	Soft Tissue 87	Lung & Bronchus 401	Esophagus 2,347	Leukemia 5,924	Leukemia 3,210
Female					
All Sites 266,693	All Sites 986	All Sites 5,664	All Sites 49,065	All Sites 129,877	All Sites 81,100
Lung & Bronchus 65,606	Leukemia 318	Breast 1,424	Breast 12,212	Lung & Bronchus 39,099	Lung & Bronchus 15,206
Breast 41,394	Brain & ONS 252	Uterine Cervix 512	Lung & Bronchus 10,886	Breast 17,405	Colon & Rectum 12,064
Colon & Rectum 28,579	Endocrine System 79	Leukemia 465	Colon & Rectum 3,891	Colon & Rectum 12,255	Breast 10,352
Pancreas 15,336	Bones & Joints 75	Lung & Bronchus 409	Ovary 3,117	Pancreas 7,722	Pancreas 5,549
Ovary 14,414	Soft Tissue 60	Colon & Rectum 364	Pancreas 1,970	Ovary 7,353	Non-Hodgkin Lymphoma 4,029

*ONS = Other nervous system.

Note: Others and Unspecified Primary are excluded from cause of death ranking order.

Source: US Mortality Public Use Data Tapes, 2001, National Center for Health Statistics, Centers for Disease Control and Prevention, 2003.

9). For all cancer sites combined, African American men have a 25% and 43% higher cancer incidence and mortality rate, respectively, than do White men. African-American women have lower incidence rates than do White females for all cancer sites combined, and yet they have 20% higher mortality rate. For the specific cancer sites listed in Table 9, incidence and mortality rates are consistently higher in African Americans than in Whites, except for breast cancer (incidence) and lung cancer (mortality) among women. Death rates from stomach and cervical cancers among African Americans are more than

twice the rates in Whites. Factors that contribute to mortality differences in African Americans and Whites include differences in exposure (eg, hepatitis C for liver cancer), access to regular screening (breast, cervical, and colorectal cancers), and timely, high-quality treatments (many cancers). The higher breast cancer incidence rates among White females are thought to reflect a combination of more frequent mammography, delayed childbearing, and perhaps greater use of hormone replacement therapy.

Among other racial and ethnic groups, cancer incidence and mortality rates are lower

TABLE 8 Trends in the Recorded Number of Cancer Deaths for Selected Cancers by Sex, US, 1989 to 2001

Year	All Sites		Lung and Bronchus		Colon and Rectum		Prostate	Breast
	Male	Female	Male	Female	Male	Female	Male	Female
1989	263,309	232,843	88,975	48,042	28,123	28,903	30,520	42,837
1990	268,283	237,039	91,014	50,136	28,484	28,674	32,378	43,391
1991	272,380	242,277	91,603	52,022	28,026	28,753	33,564	43,583
1992	274,838	245,740	91,322	54,485	28,280	28,714	34,240	43,068
1993	279,375	250,529	92,493	56,234	28,199	29,206	34,865	43,555
1994	280,465	253,845	91,825	57,535	28,471	28,936	34,902	43,644
1995	281,611	256,844	91,800	59,304	28,409	29,237	34,475	43,844
1996	281,898	257,635	91,559	60,351	27,989	28,766	34,123	43,091
1997	281,110	258,467	91,278	61,922	28,075	28,621	32,891	41,943
1998	282,065	259,467	91,399	63,075	28,024	28,950	32,203	41,737
1999	285,832	264,006	89,401	62,662	28,313	28,909	31,729	41,144
2000	286,082	267,009	90,415	65,016	28,484	28,950	31,078	41,872
2001	287,075	266,693	90,367	65,606	28,229	28,579	30,719	41,394

Note: Effective with the mortality data for 1999, causes of death are classified by ICD-10, replacing ICD-9 used for 1979 to 1998 data.

Source: US Mortality Public Use Data Tapes, 1989 to 2001, National Center for Health Statistics, Centers for Disease Control and Prevention, 2003.

for all cancer sites combined and for the four most common cancer sites than those for Whites and African Americans. However, incidence and mortality rates for cancers of the uterine cervix, stomach, and liver are generally higher in minority than in White populations. Stomach and liver cancer incidence and mortality rates are more than twice as high as in Asian/Pacific Islanders than in Whites, reflecting increased exposure to infectious agents such as *H pylori* and hepatitis C.

Historical information to adjust for delays in reporting is not available for all racial and ethnic groups. From 1992 to 2000, delay-unadjusted incidence rates for all cancer sites combined decreased by 3.1% per year among American Indians/Alaskan Natives, by 1.0% per year in African Americans, by 0.6% among Whites, and by 0.5% among Asian/Pacific Islanders, whereas incidence rates stabilized among Hispanic-Latinos (data not shown). Similarly, the death rate from all cancers combined decreased annually from 1992 through 2000 by 1.5% per year in Asian/Pacific Islanders, 1.3% among African Americans, 0.9% among Whites, and 0.3% among Hispanic-

Latinos, but it stabilized among American Indians/Alaskan Natives.³

Lifetime Probability of Developing Cancer

The lifetime probability of developing cancer is higher for men (45%) than for women (38%) (Table 10). However, because of breast cancer, women have a slightly higher probability of developing cancer before the age of 60. It is noteworthy that these estimates are based on the average experience of the general population. Thus they may overestimate or underestimate individual risk because of differences in exposure, family history, or both.

Cancer Survival Rates by Race

A poorer probability of survival once cancer is diagnosed contributes to the higher mortality rates among African-American men and women. African Americans are less likely than Whites to be diagnosed with cancer at a localized stage, when the disease may be more easily and successfully treated, and they are more likely to be di-

TABLE 9 Age-standardized Incidence and Death Rates* for Selected Cancer Sites by Race and Ethnicity, US, 1996 to 2000

	All Races	White	African American	Asian/Pacific Islander	American Indian/Alaskan Native	Hispanic-Latino†
Incidence Rates						
All Sites						
Male	555.8	555.9	696.8	392.0	259.0	419.3
Female	417.9	431.8	406.3	306.9	229.2	312.2
Breast (female)	135.0	140.8	121.7	97.2	58.0	89.8
Colon & rectum						
Male	64.2	64.1	72.4	57.2	37.5	49.8
Female	46.7	46.2	56.2	38.8	32.6	32.9
Lung & bronchus						
Male	80.8	79.4	120.4	62.1	45.6	46.1
Female	49.6	51.9	54.8	28.4	23.4	24.4
Prostate	170.1	164.3	272.1	100.0	53.6	137.2
Stomach						
Male	13.0	11.2	19.9	23.0	14.4	18.1
Female	6.2	5.1	9.9	12.8	8.3	10.0
Liver						
Male	9.0	7.3	11.0	21.1	6.1	13.8
Female	3.4	2.8	3.9	7.7	5.5	5.6
Cervix	9.6	9.2	12.4	10.2	6.9	16.8
Death Rates						
All Sites						
Male	255.5	249.5	356.2	154.8	172.3	176.7
Female	168.3	166.9	198.6	102.0	115.8	112.4
Breast (female)	27.7	27.2	35.9	12.5	14.9	17.9
Colon & rectum						
Male	25.8	25.3	34.6	15.8	18.5	18.4
Female	18.0	17.5	24.6	11.0	12.1	11.4
Lung & bronchus						
Male	79.5	78.1	107.0	40.9	52.9	40.7
Female	40.7	41.5	40.0	19.1	26.2	15.1
Prostate	32.9	30.2	73.0	13.9	21.9	24.1
Stomach						
Male	6.9	6.1	14.0	12.5	7.0	9.9
Female	3.4	2.9	6.5	7.4	4.2	5.3
Liver						
Male	6.6	6.0	9.3	16.1	7.6	10.5
Female	2.9	2.7	3.7	6.7	4.3	5.0
Cervix	3.0	2.7	5.9	2.9	2.9	3.7

*Rates are per 100,000 and age-adjusted to the 2000 US standard population.

†Hispanic-Latino are not mutually exclusive from Whites, African Americans, Asian/Pacific Islanders and American Indians/Alaskan Natives.

Source: Ries LAG, Eisner MP, Kosary CL, et al.³

TABLE 10 Probability of Developing Invasive Cancers Within Selected Age Intervals, by Sex, US, 1998 to 2000*

		Birth to 39	40 to 59	60 to 79	Birth to Death
		(%)	(%)	(%)	(%)
All Sites †	Male	1.36 (1 in 73)	8.03 (1 in 12)	33.92 (1 in 3)	44.77 (1 in 2)
	Female	1.92 (1 in 52)	9.01 (1 in 11)	22.61 (1 in 4)	38.03 (1 in 3)
Bladder ‡	Male	.02 (1 in 4603)	.40 (1 in 250)	2.36 (1 in 42)	3.46 (1 in 29)
	Female	.01 (1 in 9557)	.12 (1 in 831)	.64 (1 in 157)	1.10 (1 in 91)
Breast	Female	.44 (1 in 229)	4.14 (1 in 24)	7.53 (1 in 13)	13.36 (1 in 7)
Colon & Rectum	Male	.06 (1 in 1678)	.86 (1 in 116)	3.94 (1 in 25)	5.88 (1 in 17)
	Female	.06 (1 in 1651)	.67 (1 in 150)	3.05 (1 in 33)	5.49 (1 in 18)
Leukemia	Male	.15 (1 in 649)	.20 (1 in 495)	.82 (1 in 122)	1.45 (1 in 70)
	Female	.13 (1 in 789)	.14 (1 in 706)	.46 (1 in 219)	1.00 (1 in 100)
Lung & Bronchus	Male	.03 (1 in 3439)	1.02 (1 in 98)	5.80 (1 in 17)	7.69 (1 in 13)
	Female	.03 (1 in 3046)	.79 (1 in 126)	3.93 (1 in 25)	5.73 (1 in 17)
Melanoma of the Skin	Male	.12 (1 in 809)	.49 (1 in 205)	.97 (1 in 103)	1.81 (1 in 55)
	Female	.19 (1 in 532)	.39 (1 in 255)	.51 (1 in 197)	1.22 (1 in 82)
Non-Hodgkin Lymphoma	Male	.14 (1 in 739)	.45 (1 in 224)	1.27 (1 in 79)	2.10 (1 in 48)
	Female	.08 (1 in 1258)	.30 (1 in 332)	.98 (1 in 102)	1.76 (1 in 57)
Prostate	Male	.01 (1 in 12833)	2.28 (1 in 44)	14.20 (1 in 7)	17.15 (1 in 6)
Uterine Cervix	Female	.16 (1 in 632)	.31 (1 in 322)	.27 (1 in 368)	.78 (1 in 128)
Uterine Corpus	Female	.05 (1 in 1832)	.69 (1 in 144)	1.57 (1 in 64)	2.60 (1 in 38)

* For those free of cancer at beginning of age interval. Based on cancer cases diagnosed during 1998 to 2000.

† The "1 in" statistic and the inverse of the percentage may not be equivalent due to rounding.

‡ All Sites exclude basal and squamous cell skin cancers and in situ cancers except urinary bladder.

Source: DEVCAN Software, Probability of Developing or Dying of Cancer Software, Version 5.1. Statistical Research and Applications Branch, National Cancer Institute, 2003. <http://srab.cancer.gov/devacan>.

agnosed with cancer at a regional or distant stage. This is true for most of the common cancer sites (Figure 7). Furthermore, for nearly every cancer site, African Americans have lower five-year relative survival rates than do Whites at each stage of diagnosis (Figure 8), suggesting the possible influences of disparities in access and receipt of quality health care and of differences in comorbid conditions. The extent to which these factors, individually or collectively, contribute to the overall differential survival rate is not entirely clear.¹⁸ However, recent findings suggest that African Americans and Whites have similar disease outcomes when they receive similar cancer treatment and medical care.¹⁹

The relative five-year survival rates for most of the selected cancer sites and all

cancer sites combined have improved notably over time (Table 11). This is true for both Whites and African Americans. Cancer sites without significant improvement in survival rates in the past 25 years include the uterine corpus, cervix, larynx, liver, lung, pancreas, stomach, and esophagus. For nearly all cancer sites, survival rates for African Americans remain substantially lower than survival rates for Whites.

The relative survival rate cannot be calculated for other racial and ethnic groups because accurate life expectancy data for these population subgroups are not available. However, based on cause-specific survival rates of persons with cancer from 1988 to 1997 in the SEER areas, most minority pop-

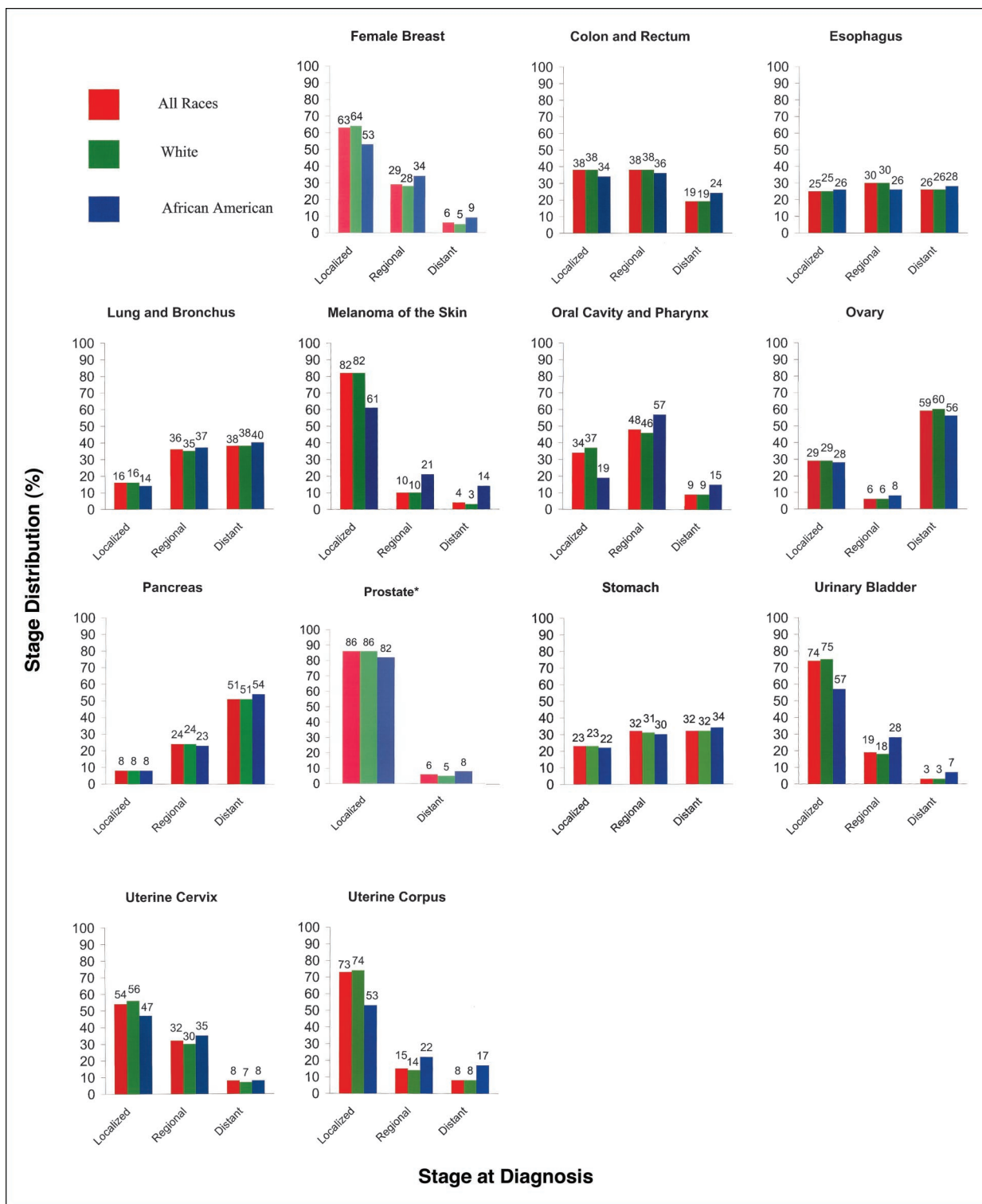


FIGURE 7 Distribution of Cancer Cases for Selected Cancer Types, by Race and Stage at Diagnosis, US, 1992 to 1999

*The rate for localized stage represents localized and regional stages combined.

Note: Staging according to Surveillance, Epidemiology, and End Results (SEER) historic stage categories rather than the American Joint Committee on Cancer (AJCC) staging system. For each type and race, stage categories do not total 100% because sufficient information is not available to assign a stage to all cancer cases.

Source: Ries LAG, Eisner MP, Kosary CL, et al.⁹

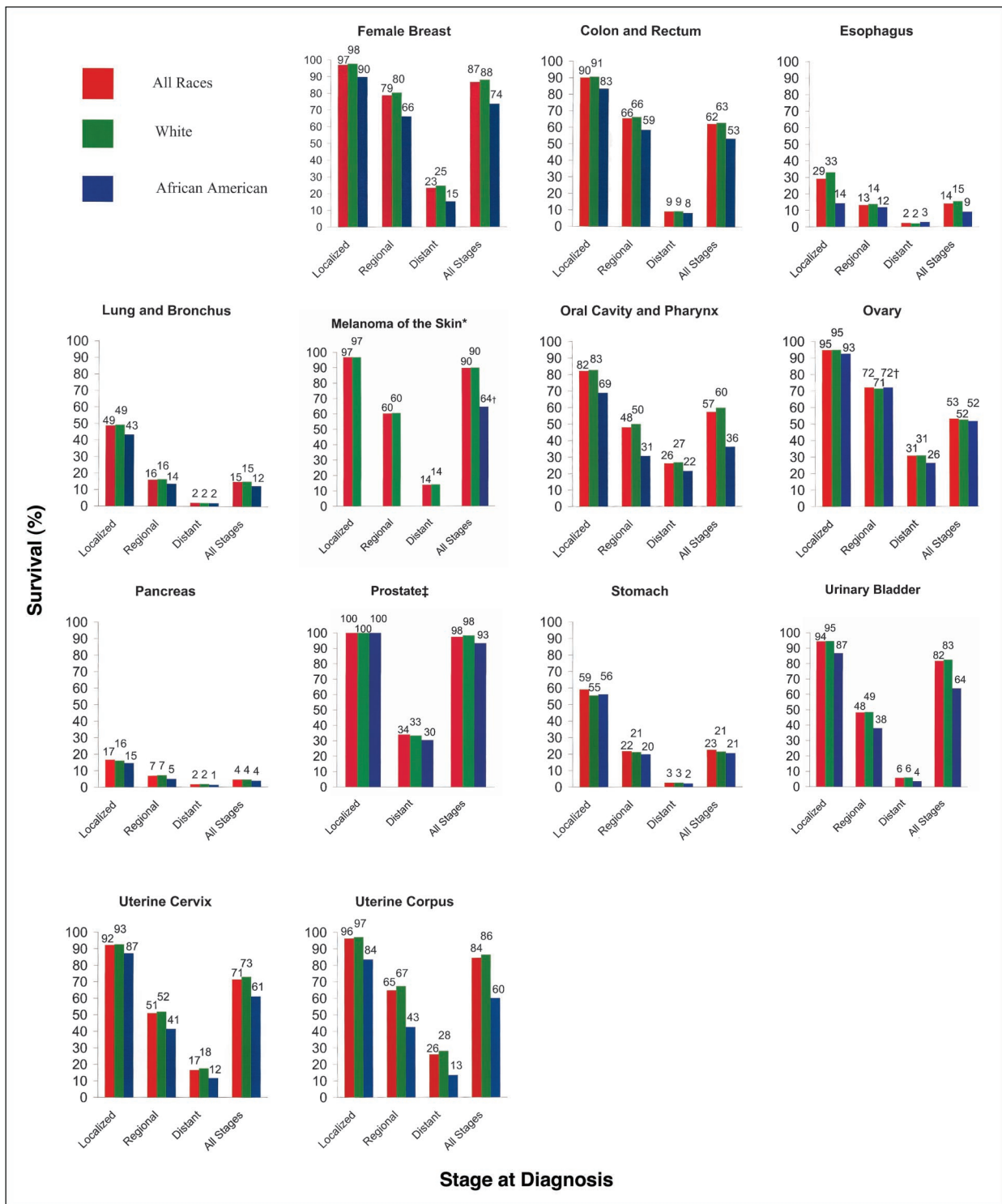


FIGURE 8 Five-year Relative Survival Rates Among Patients Diagnosed with Selected Cancer Types, by Race and Stage at Diagnosis, US, 1992 to 1999

*Data for localized, regional, and distant stage melanoma of the skin for African American is not shown.

†The standard error is between 5 and 10 percentage points.

‡The rate for localized stage represents localized and regional stages combined.

Note: Staging according to Surveillance, Epidemiology, and End Results (SEER) historic stage categories rather than the American Joint Committee on Cancer (AJCC) staging system.

Source: Source: Ries LAG, Eisner MP, Kosary CL, et al.³

TABLE 11 Trends in Five-year Relative Survival Rates* (%) for Selected Cancer Sites by Race and Year of Diagnosis, US, 1974 to 1999

Site	Relative Five-Year Survival Rate (%)								
	White			African American			All Races		
	1974 to 1976	1983 to 1985	1992 to 1999	1974 to 1976	1983 to 1985	1992 to 1999	1974 to 1976	1983 to 1985	1992 to 1999
All cancers	51	54	64 †	39	40	53 †	50	53	63 †
Brain	22	26	32 †	27	32	39 †	22	27	33 †
Breast (female)	75	79	88 †	63	64	74 †	75	78	87 †
Cervix uterine	70	71	73 †	64	61	61	69	69	71 †
Colon	51	58	63 †	46	49	53 †	50	58	62 †
Uterine corpus	89	85	86 †	62	55	60	88	83	84 †
Esophagus	5	9	15 †	4	6	9 †	5	8	14 †
Hodgkin disease	72	79	85 †	69	77	78 †	71	79	84 †
Kidney	52	56	63 †	49	55	61 †	52	56	63 †
Larynx	66	69	67	60	55	53	66	67	65
Leukemia	35	42	48 †	31	34	39	34	41	46 †
Liver	4	6	7 †	1	4	5 †	4	6	7 †
Lung & bronchus	13	14	15 †	11	11	12 †	13	14	15 †
Melanoma of the skin	81	85	90 †	67 ‡	74 §	64 ‡	80	85	90 †
Multiple myeloma	24	27	31 †	28	31	33	24	28	32 †
Non-Hodgkin lymphoma	48	54	57 †	49	45	47	47	54	56 †
Oral cavity	55	55	60 †	36	35	36	54	53	57 †
Ovary	37	40	52 †	41	42	52 †	37	41	53 †
Pancreas	3	3	4 †	3	5	4	3	3	4 †
Prostate	68	76	98 †	58	64	93 †	67	75	98 †
Rectum	49	56	62 †	42	44	53 †	49	55	62 †
Stomach	15	16	21 †	17	19	21	15	17	23 †
Testis	79	91	96 †	76 ‡	88 ‡	87	79	91	96 †
Thyroid	92	93	96 †	88	92	94	92	94	96 †
Urinary bladder	74	78	83 †	48	60	64 †	73	78	82 †

* Survival rates are adjusted for normal life expectancy and are based on cases diagnosed from 1974 to 76, 1983 to 85, and 1992 to 99 and followed through 2000.

† The difference in rates between 1974 to 76 and 1992 to 1999 is statistically significant ($p < 0.05$).

‡ The standard error of the survival rate is between 5 and 10 percentage points.

§ The standard error of the survival rate is greater than 10 percentage points.

Source: Ries LAG, Eisner MP, Kosary CL, et al.³

ulations have an elevated probability of dying of cancer within five years of diagnosis, compared with non-Hispanic Whites after accounting for differences in age and stage at diagnosis.²⁰ For most cancers, minority populations are also less likely to be diagnosed at a localized stage and more likely to be diagnosed at distant stage, compared with White populations. Based on all solid tumors diagnosed with stage information from 1996 to 2000 in the SEER areas, 30% of the cases

among American Indians/Alaskan Natives, 26% in African Americans and Hispanic-Latinos, and 25% among Asian/Pacific Islanders were diagnosed at distant stage, compared with 22% in Whites.^{4,5}

Cancer in Children

Cancer is the second leading cause of death among children between the ages of 1 and 14 years in the United States; accidents

TABLE 12 Fifteen Leading Causes of Death Among Children Ages 1 to 14, US, 2001

Rank	Cause of Death	Number of Deaths	Percent (%) of Total Deaths	Death Rate*
	All Causes	12,202	100.0	21.7
1	Accidents (Unintentional Injuries)	4,550	37.3	8.1
2	Cancer	1,428	11.7	2.5
3	Congenital Anomalies	933	7.6	1.7
4	Assault (Homicide)	734	6.0	1.3
5	Heart Diseases	497	4.1	0.9
6	Intentional Self-harm (Suicide)	279	2.3	0.5
7	Pneumonia & Influenza	204	1.7	0.4
8	Septicemia	177	1.5	0.3
9	In situ/Benign/Unknown Neoplasms	163	1.3	0.3
10	Chronic Lower Respiratory Disease	147	1.2	0.3
11	Cerebrovascular Disease	134	1.1	0.2
12	Anemias	77	0.6	0.1
13	HIV Disease	51	0.4	0.1
14	Meningitis	50	0.4	0.1
15	Diabetes Mellitus	45	0.4	0.1
	All Other Causes	2,733	22.4	

Percentages may not total 100 due to rounding. Certain Perinatal Condition was excluded from ranking order.

*Rates are per 100,000 population and age-adjusted to the 2000 US standard population.

Source: US mortality Public Use Data Tape, 2001, National Center for Health Statistics, Centers Disease Control and Prevention, 2003.

TABLE 13 Trends in Five-year Relative Cancer Survival Rates* (%) for Children Under Age 15, US, 1974 to 1999

Site	Five Year Relative Survival Rates (%)						
	Year of Diagnosis						
	1974 to 1976	1977 to 1979	1980 to 1982	1983 to 1985	1986 to 1988	1989 to 1991	1992 to 1999
All Sites	56	62	65	68	70	73	78 †
Acute Lymphocytic Leukemia	53	67	71	69	78	80	85 †
Acute Myeloid Leukemia	14	27 ‡	24 ‡	29 ‡	30 ‡	35 ‡	47 †
Bones and Joints	55 ‡	52 ‡	55 ‡	57	63 ‡	62	72 †
Brain and Other Nervous System	55	56	56	62	63	62	70 †
Hodgkin Disease	78	84	91	90	90	94	94 †
Neuroblastoma	53	53	53	55	59	68	68 †
Non-Hodgkin Lymphoma	44	50	62	71	70	75	81 †
Soft Tissue	60	68	65	76	67	78	74 †
Wilm's Tumor	74	78	87	87	91	93	91 †

* Survival rates are adjusted for normal life expectancy and are based on follow-up of patients through 2000.

† The difference in rates between 1974 to 1976 and 1992 to 1999 is statistically significant ($p < 0.05$).

‡ The standard error of the survival rate is between 5 and 10 percentage points.

Note: "All Sites" excludes basal and squamous cell skin cancers and in situ carcinomas except urinary bladder.

Source: Ries LAG, Eisner MP, Kosary CL, et al.³

are the most frequent cause of death in this age group (Table 12). The most commonly occurring cancers found in children (0 to 14 years) are leukemia (particularly acute lymphocytic leukemia), tumors of the central and sympathetic nervous systems, lymphoma, soft tissue sarcomas, and renal tumors.³ In the past 25 years, the five-year relative survival rate has improved significantly for many childhood cancers, including non-Hodgkin lymphoma, acute lymphocytic leukemia, acute myeloid leukemia, and Wilms' tumor (Table 13). The five-year relative survival rate among children for all cancer sites combined improved from 56% for patients diagnosed in 1974 to 1976 to 78% for those diagnosed in 1992 to 1999.³

STUDY LIMITATIONS AND FUTURE CHALLENGES

Estimates of the expected numbers of new cancer cases and cancer deaths should be interpreted with caution. These estimates may vary considerably from year to year, particularly for

less common cancers and in states with smaller populations. Unanticipated changes may occur that are not captured by our modeling effort. The estimates of new cancer cases are based on incidence rates for the geographic locations that participate in the SEER program and, therefore, may not be representative of the entire United States. For these reasons, we discourage the use of estimates to track year-to-year changes in cancer occurrence and death. The recorded number of cancer deaths and cancer mortality rates from the National Center for Health Statistics and cancer incidence rates from SEER are generally the preferred data sources for tracking cancer trends, even though these data are three and four years old, respectively, when the estimates are calculated.

Despite these limitations, the American Cancer Society estimates do provide evidence of current patterns of cancer incidence and death in the United States. Such estimates will assist in continuing efforts to reduce the public health burden of cancer.

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